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(54) **TONG WITH ONE-HANDED LOCKING MECHANISM**

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USPC 294/99.2
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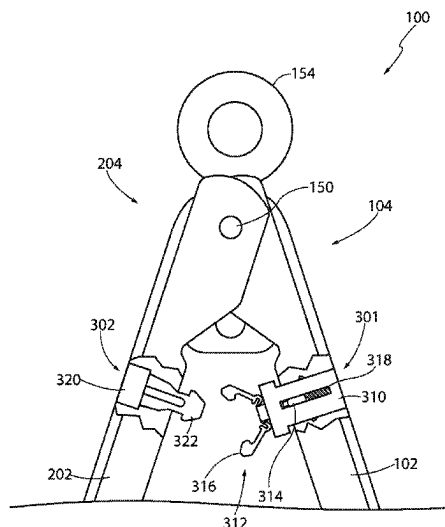
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(57) **ABSTRACT**

A tong with a lock that can be actuated using one hand without changing a grip on the tong and without having to change an orientation of the tong. The lock can be engaged and disengaged by squeezing arms of the tong together. The lock can have a latch on one arm and a strike on the other arm. When the arms are squeezed together the strike engages the latch into a locked configuration. From the locked configuration, when the arms are squeezed again, the latch releases the strike.

8 Claims, 5 Drawing Sheets



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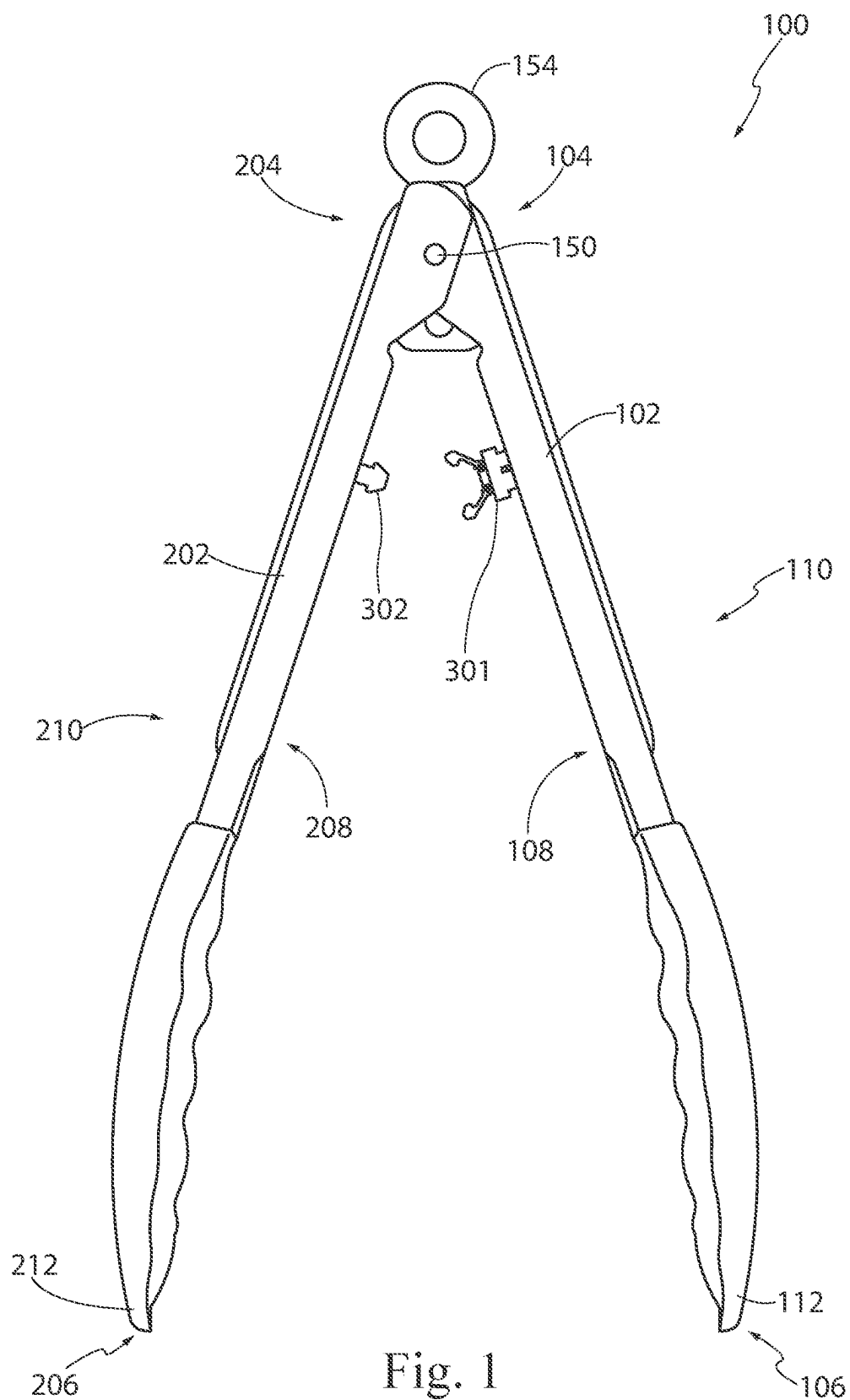


Fig. 1

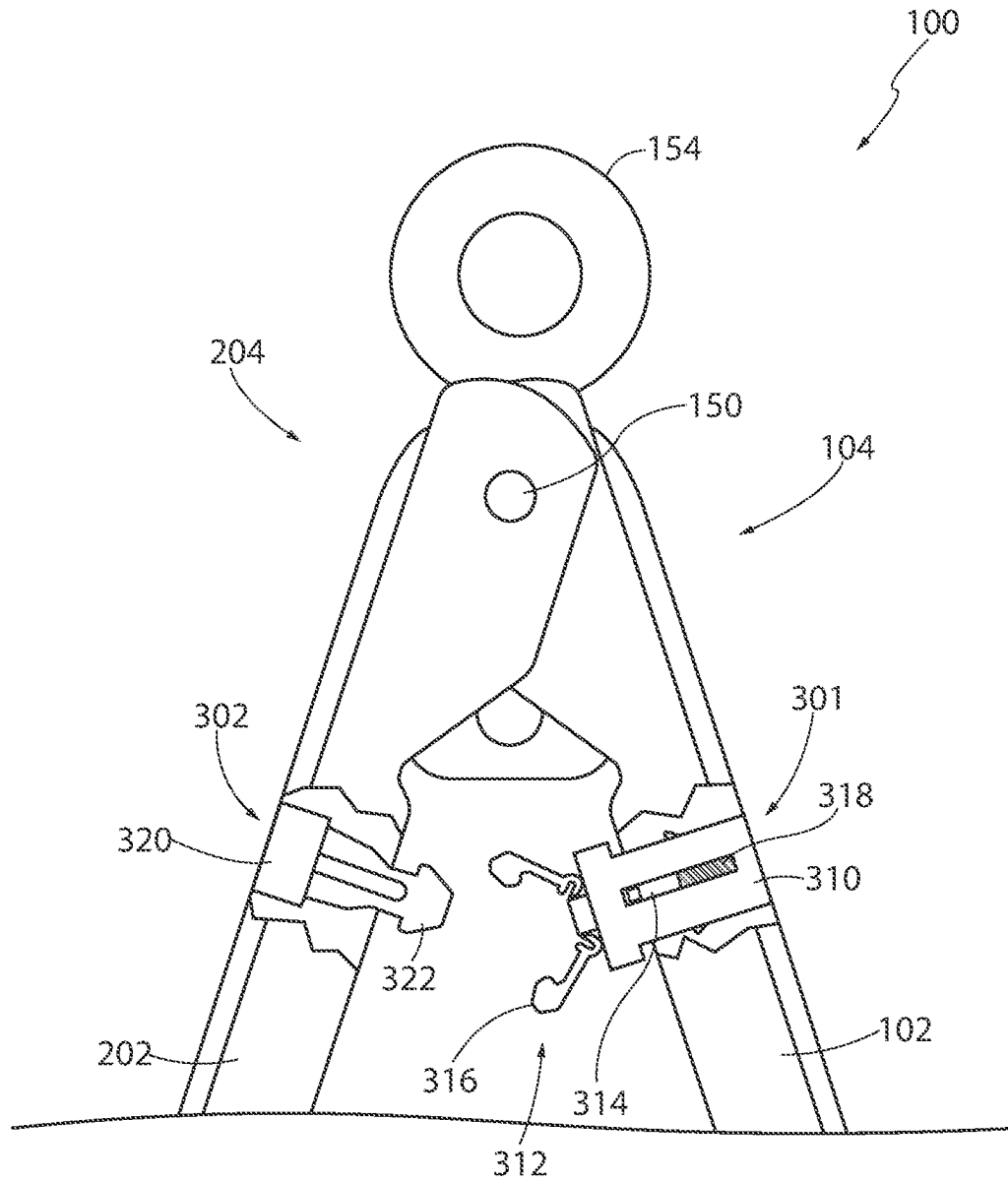


Fig. 2

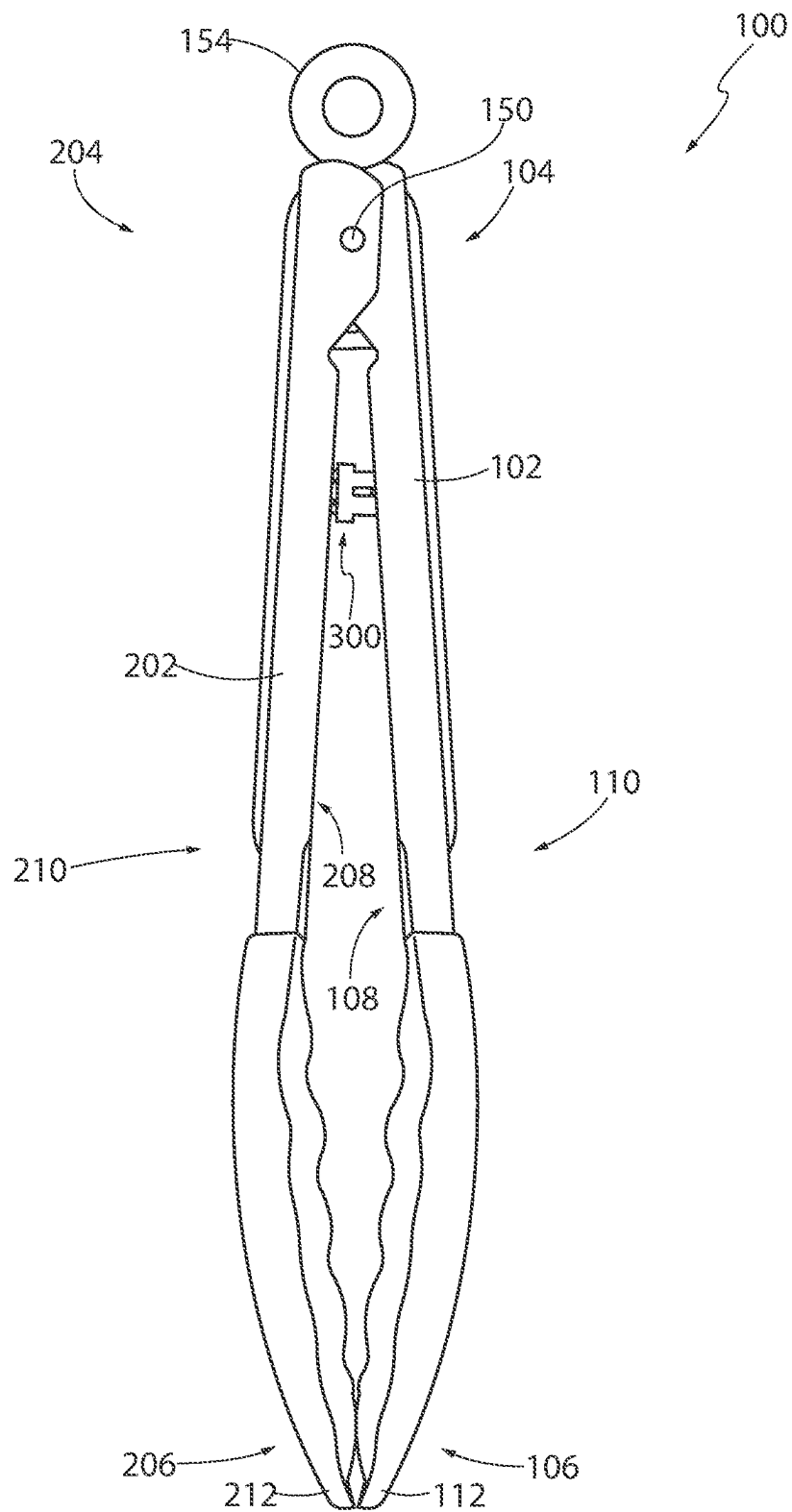


Fig. 3

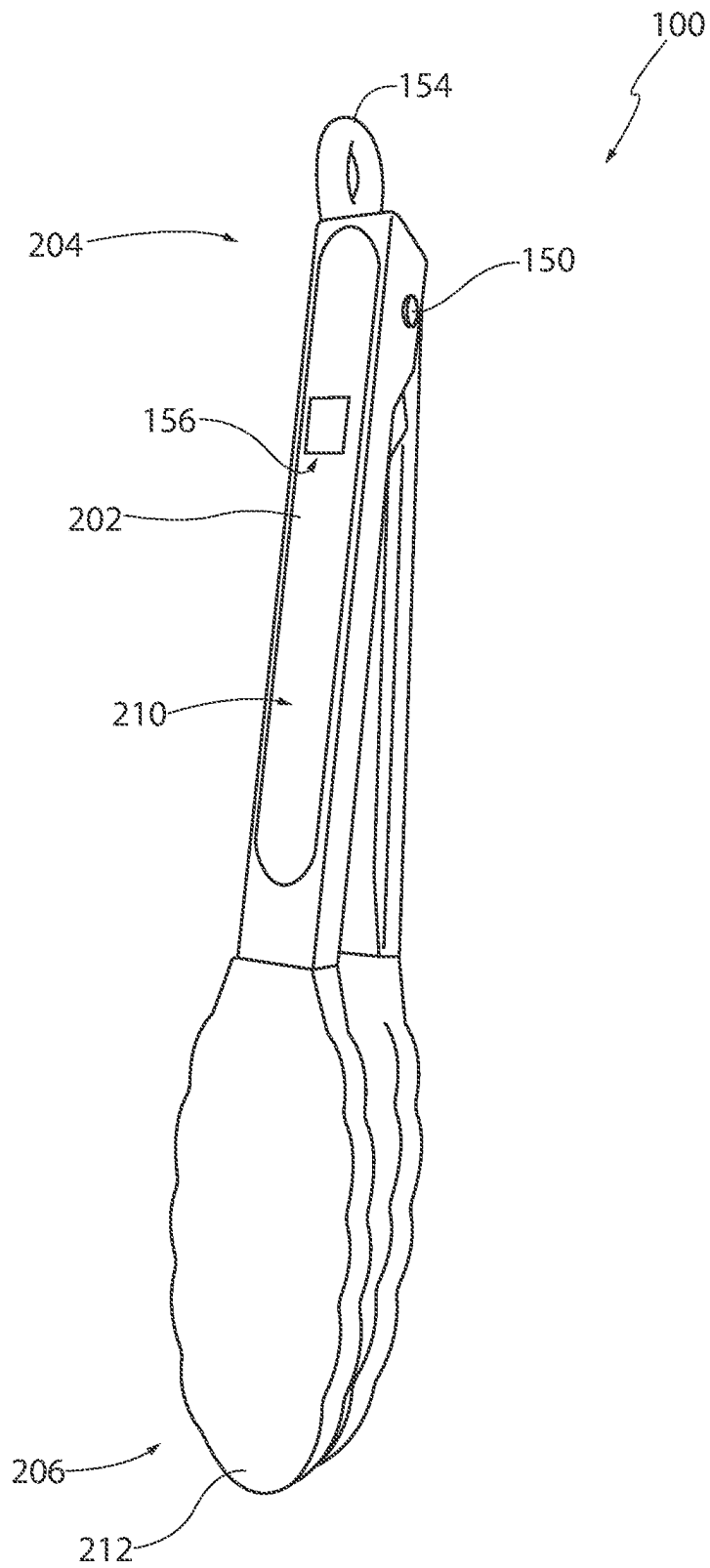


Fig. 4

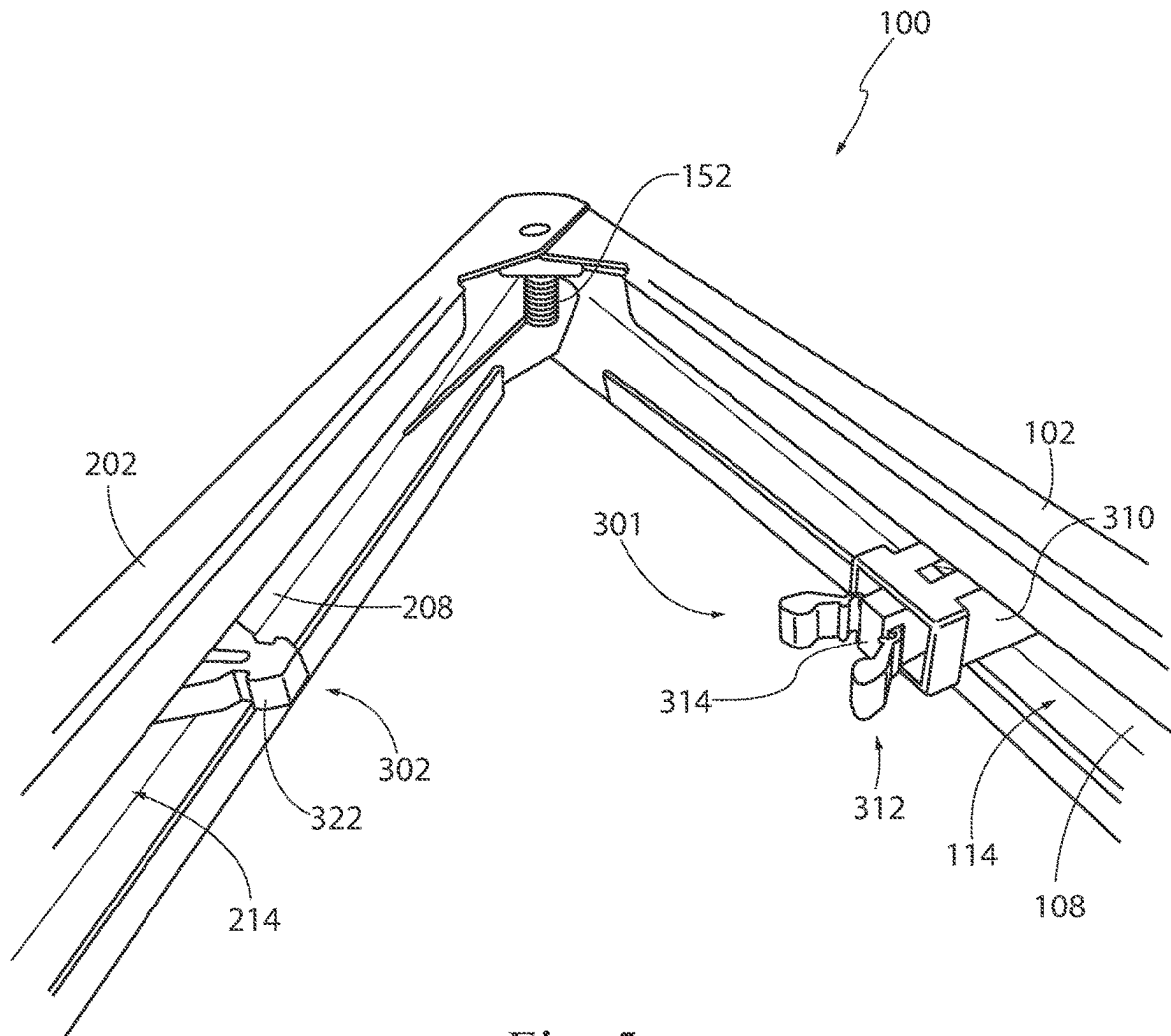


Fig. 5

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TONG WITH ONE-HANDED LOCKING MECHANISM**CROSS REFERENCE TO RELATED APPLICATIONS**

This patent application is a continuation of PCT Application No. PCT/US2022/076661 filed on Sep. 29, 2022, which is a continuation of U.S. patent application Ser. No. 17/495,567, filed Oct. 6, 2021, which applications are incorporated in their entirety here by this reference.

TECHNICAL FIELD

This invention relates to tongs, such as those used in in cooking.

BACKGROUND

Tongs are a very common cooking utensil used in the kitchen and at barbecues allowing the cook to pick up food items without being too close to the heat. The tong allows the cook to add food into the hot cooking area, flip foods, remove food, and the like. Generally, the tong has two arms connected by a spring-loaded hinge at one end allowing the free end to be squeezed to grasp the food item.

Oftentimes it is convenient to have the tong arms locked in place, particularly, in the closed configuration. The most popular locking mechanism is the use of a pull tab located at the hinge where the first arm is connected to the second arm. In order to lock the tong in the closed configuration, the cook closes the tong with one hand and pulls on the tab with a free hand. To unlock the tong, the cook holds the tong in one hand and pushes the tab back towards the hinge. Thus, this common locking mechanism requires both hands to lock and unlock the tong arms. Other locks require the user to shift his or her grip to access the lock.

To overcome the need for two hands or shifting grips to lock and unlock the tong arms, another popular mechanism is the use of a gravity-based lock. With the gravity-based locks, the user squeezes the tongs together and the user holds the distal ends upward or downward to allow gravity to pull the lock into a locked or unlocked configuration. These tongs require the added motion of changing the orientation of the tongs, which may not always be desirable. For example, if there are liquids, oils, or sauces on the distal end of the tong, the user may not want to hold the distal end upward and allow the liquid, oil, or sauce to drip towards the hinge.

For the foregoing reasons there is a need for a tong that can be locked or unlocked with a single hand, without changing the grip on the tong, and without having to change the orientation of the tong.

DISCLOSURE OF INVENTION

The present invention is directed to a tong that can be locked or unlocked with a single hand, without changing the grip on the tong, and without having to change the orientation of the tong. The tong utilizes a locking mechanism positioned on the interior surface of the arms that can lock together simply by squeezing the arms together, and unlock simply by squeezing the arms together.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 shows a side elevation view of an embodiment of the present invention in the open and unlocked configuration;

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FIG. 2 shows a close up of an embodiment of the hinge and locking mechanism with a portion of the arms removed to see inside the arms;

FIG. 3 shows the embodiment of FIG. 1 in the closed and locked configuration.

FIG. 4 shows a perspective view of an embodiment of the present invention; and

FIG. 5 shows a perspective view of a portion of an embodiment of the present invention from the interior side.

MODES OF CARRYING OUT THE INVENTION

The detailed description set forth below in connection with the appended drawings is intended as a description of presently-preferred embodiments of the invention and is not intended to represent the only forms in which the present invention may be constructed or utilized. The description sets forth the functions and the sequence of steps for constructing and operating the invention in connection with the illustrated embodiments. It is to be understood, however, that the same or equivalent functions and sequences may be accomplished by different embodiments that are also intended to be encompassed within the spirit and scope of the invention.

With reference to FIGS. 1-5, the invention of the present invention is a tong 100 having two arms 102, 202, and a lock 300 that can be engaged to lock the arms 102, 202 in place, and disengaged to allow the arms 102, 202 to move, with one hand and without having to move the user's finger positioning or change the orientation of the tong 100, simply by squeezing the arms 102, 202 together. As such, the user can use the tong 100 as intended, and when the user wants to lock the arms 102, 202, the user can simply squeeze the tong 100 with nothing in between the arms 102, 202, and disengage the lock, again without adjusting the user's grips or changing the tong 100 orientation, simply by squeezing the arms 102, 202 together. Thus, the same action (squeezing the arms 102, 202) for locking the arms 102, 202 is the same action for unlocking the arms 102, 202.

The tong 100 comprises a first arm 102 having a first proximal end 104, a first distal end 106 opposite the first proximal end 104, a first inner surface 108 in between the first proximal end 104 and the first distal end 106, and a first outer surface 110 opposite the first inner surface 108 and in between the first proximal end 104 and the first distal end 106. The tong 100 also comprises a second arm 202 having a second proximal end 204, a second distal end 206 opposite the second proximal end 204, a second inner surface 208 in between the second proximal end 204 and the second distal end 206, and a second outer surface 210 opposite the second inner surface 208 and in between the second proximal end 204 and the second distal end 206. The first arm 102 is connected to the second arm 202 by a hinge 150 at the first and second proximal ends 104, 204.

The arms 102, 202 can be made of a rigid material, such as metal, plastic, wood, and the like. In the preferred embodiment, the arms 102, 202 are generally straight such that when the first distal end 106 touches the second distal end 206, the first arm 102 and the second arm 202 are substantially parallel as shown in FIGS. 3 and 4. For example, as measured at the proximal ends 104, 204, when the first distal end 106 is in contact with the second distal end 206, the first arm 102 and the second arm 202 may form an angle of about 15 degrees or less. More preferably, the angle may be about 10 degrees or less. Most preferably, the angle may be about 5 degrees or less.

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The first proximal end **104** of the first arm **102** and the second proximal end **204** of the second arm **202** are connected by a hinge **150**. The hinge **150** allows the distal ends **106**, **206** to move towards and away from each other to allow the user to pick up objects at the distal ends **106**, **206**. As shown in FIG. 5, a tong spring **152** can be placed at the hinge **150** imposing a biasing force against the first arm **102** and the second arm **202** causing the first distal end **106** to remain separated and distal from the second distal end **206** as shown in FIG. 1. In the preferred embodiment, the first distal end **106** can have a first flexible tip **112**, and the second distal end **206** can have a second flexible tip **212**. The flexible tips **112**, **212** can be made of a durable yet flexible material, such as silicone, plastic, rubber, and the like.

The flexibility of the tips **112**, **212** facilitates in the locking mechanism. Specifically, the lock **300** is mounted on the first arm **102** and the second arm **202** such that in order for the lock **300** to engage, the first arm **102** and the second arm **202** must be closer to each other than they are when the first flexible tip **112** just touches the second flexible tip **212**. Thus, when the first flexible tip **112** just touches the second flexible tip **212**, the first and second arms are not locked together. Rather, the arms **102**, **202** must over-travel causing the flexible tips **112**, **212** to bend slightly to allow the arms **102**, **202** to come even closer to each other to allow the lock **300** to engage as shown in FIG. 3 and lock the arms **102**, **202** together. This over-travel feature reduces the probability of the first and second arms **102**, **202** inadvertently locking during regular use. In other words, when picking up a small or thin piece of material with the tongs **100**, the user would not want the tongs **100** to lock inadvertently because the first arm **102** is too close to the second arm **202**.

In embodiments with the flexible tips **112**, **212**, the first inner surface **108** of the first arm **102** can define a first channel **114**, and the second inner surface **208** of the second arm **202** can define a second channel **214** as shown in FIG. 5. The first channel **114** and the second channel **214** provide depth for the first and second arms **102**, respectively, to provide space to mount the first flexible tip **112** and second flexible tip **212** on to the first arm **102** and second arm **202**, respectively.

In the preferred embodiment, the locking mechanism can be in the form of a latch **301** and strike **302**, such as the grab catch sold by Grainger. The latch **301** can be mounted on the first inner surface **108** of the first arm **102**. The strike **302** can be mounted on the second inner surface **208** of the second arm **202**. The latch **301** and strike **302** are configured to connect to each other when the first arm **102** and the second arm **202** are adjacent to each other.

In the preferred embodiment, the lock **300** comprises a latch **301** and a strike **302**. In the engaged and locked configuration, the latch **301** receives and engages the strike **302** to prevent the strike from being removed from the latch. With reference to FIG. 2, the latch **301** can comprise a housing **310** mounted on the first inner surface **108** and a spring-loaded catch **312** mounted on the housing **310**. The spring-loaded catch **312** can have a post **314** and a clasp **316** connected to the post **314**. A spring **318** inside the housing can bias against the post **314**.

The strike **302** can comprise a base **320** attached to the second inner surface **208** and a flanged head **322** opposite the base **320**. The latch **301** and strike **302** are configured to connect to each other in a locked configuration when the first arm **102** and the second arm **202** are substantially parallel to each other. In the engaged and locked configuration, the flanged head **322** abuts against the post **314** with the clasp **316** surrounding the flanged head **322** as shown in FIG. 3.

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In the preferred embodiment, when the strike **302** is pressed against the post **314** of the latch **301**, the strike **302** pushes the post **314** deeper into the housing **310**. When the post **314** descends into the housing **310** it draws the clasp **316** together causing the clasp **316** to close around the flanged head **322** as the clasp **316** and the flanged head **322** enter into the housing **310**. Inside the housing **310** is a latch spring **318** abutting against the post **314** creating a biasing force against the post **314** to push the post **314** out of the housing **310**. When the post **314** is pushed deep enough into the housing **310** an internal catch (not shown) locks the post **314** in place, and when the squeezing pressure is released, the arms **102**, **202** are locked in place. Squeezing the arms **102**, **202** again causes the internal catch to release and the latch spring **318** pushes the post **314** outwardly causing the clasp **316** to emerge from the housing **310** and open, which releases the strike **302**, while the tong spring **152** causes the distal ends **106**, **206** of the arms **102**, **202** to separate.

In the preferred embodiment, the latch **301** and the strike **302** are positioned on the first and second arms **102**, **202** in a manner that when in the locked configuration with the strike **302** engaged with the latch **301**, the first distal end **106** touches the second distal end **206**. In embodiments with the first and second flexible tips **112**, **212**, the first and second flexible tips **112**, **212** are the distal ends **106**, **206**, and the first and second flexible tips **112**, **212** touch each other in the locked configuration at their free ends.

In the preferred embodiment, in the locked configuration, the latch **301** and the strike **302** are entirely interior to the first outer surface **110** and the second outer surface **210**. In this configuration, because the latch **301** and the strike **302** are located interiorly, the latch **301** and strike **302** are unlikely to be touched during normal use when the tong **100** is held in a natural and comfortable manner. When the tong **100** is handled naturally and properly, the user's hands only make contact with the first and second outer surfaces **110**, **210** and the sides of the tong **100**, and not the interior surfaces **108**, **208**. As such, unlike other tongs in which the locking mechanism is exposed on the exterior surface, the tong **100** of the present invention has no chance for the user to inadvertently disengage the lock **300** by inadvertently touching any part of the locking mechanism.

In the preferred embodiment, the latch **301** and the strike **312** are configured along the first and second arms **102**, **202**, respectively, to lock only when the first flexible tip **106** is compressed against the second flexible tip **206**. In other words, just be the nature of the first flexible tip **106** touching the second flexible tip **206** is not sufficient to cause the latch **301** and the strike **302** to lock. Rather, sufficient squeezing force must be applied causing the first flexible tip **106** and the second flexible tip **206** to flex or bend away from each other before the latch **301** catches the strike **302**. This configuration reduces the chances of the latch **301** catching the strike **302** during normal use when the user does not intend to lock the tong **100**.

To facilitate locking the arms **102**, **202** at the proper positioning, the latch **301** can reside inside the first channel **114**, and the strike **302** can reside inside the second channel **214**. Preferably, a portion of the latch **301** can protrude out from the first channel **114** directed interiorly towards the strike **302**, and a portion of the strike **302** can protrude out from the second channel **214** directed interiorly towards the latch **301**. Preferably, the portion of the latch **301** that protrudes out from the first channel **114** is at least the clasp **312**, and the portion of the strike **302** that protrudes out from the second channel **214** is the flanged head **322**. Because the latch **301** and the strike **302** protrude interiorly, no portion

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of the latch **301** or the strike **302** protrudes from the exterior surface of the arms **102**, **202** where the locking mechanism can be accidentally dislodged.

In the preferred embodiment, the latch **301** and strike **302** are configured to disconnect from each other from the locked configuration when the first arm **102** and second arm **202** are squeezed towards each other. Therefore, if the user locks the tong **100** in the closed configuration and keeps the tong **100** in the user's hands, in order to unlock the arms **102**, **202** while maintaining the same grip, the user simply has to squeeze the arms **102**, **202** together a little closer and the latch **301** releases the strike **302**. Thus, there is no need to shift the positioning of the hands on the tong **100**, there is no need to look for a release button, there is no need for the user to use his or her opposite hand to release the lock, there is no need to change the orientation of the tong **100**, and the like. The present invention allows the tong **100** to lock and unlock using only the natural movement for using a tong, namely, squeezing the arms **102**, **202**.

In the preferred embodiment, the latch **301** and the strike **302** are fixed along the first arm **102** and second arm **202**, respectively. In some embodiments, the latch **301** and the strike **302** can be adjustable along the length of the first and second arms **102**, **202**. Stops may be placed at set positions along the first and second arms **102**, **202** so that the latch **301** and the strike **302** can be properly aligned. In the preferred embodiment, the lock **300** is positioned towards the middle portion of the arms **102**, **202**. For example, the lock **300** may be positioned at least approximately 2 inches from the hinge **150** towards the distal ends **106**, **206**. In some embodiments, the lock **300** may be at least approximately 3 inches from the hinge **150** towards the distal ends **106**, **206**. In some embodiments, the lock **300** may be at least approximately 4 inches from the hinge **150** towards the distal ends **106**, **206**. In some embodiments, the lock **300** may be at least approximately 5 inches from the hinge **150** towards the distal ends **106**, **206**. In some embodiments, the lock **300** may be at least approximately 6 inches from the hinge **150** towards the distal ends **106**, **206**. In some embodiments, the lock **300** may be at least approximately 7 inches from the hinge **150** towards the distal ends **106**, **206**. For longer tongs **100**, the lock **300** can be even farther away from the hinge **150** and closer to the distal ends **106**, **206**. The lock **300** can be fastened to the arms **102**, **202** using any type of fastening mechanism, such as adhesives, screws, welding, tongue and groove, and the like. In some embodiments, an opening **156** may have to be created through the arms **102**, **202** to accommodate the lock **300**.

In some embodiments, the tong **100** can further comprise a secondary lock **154** adjacent to the joint **150** to assist in keeping the arms **102**, **202** in the locked configuration. For example, if the user does not intend on using the tong **100** for a prolonged period of time, to reduce the stress of the tong spring **152** on the latch **301** and strike **302**, the secondary lock **154** can be employed to keep the first arm **102** and second arm **202** together. The secondary lock **154** can prolong the life of the latch **301** and strike **302**.

In use, the user uses the tong **100** as tongs are normally intended. When the user wants to lock the arms **102**, **202** in place, the user squeezes the arms **102**, **202** together compressing the first flexible tip **112** against the second flexible tip **212** to lock the latch **301** with the strike **302**. To unlock the tong **100**, without moving a position of any fingers on the first and second arms **102**, **202**, the user squeezes the first arm **102** towards the second arm **202** to release the latch **301** from the strike **302**.

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The foregoing description of the preferred embodiment of the invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the invention not be limited by this detailed description, but by the claims and the equivalents to the claims appended hereto.

INDUSTRIAL APPLICABILITY

This invention may be industrially applied to the development, manufacture, and use of tongs, generally used as a kitchen appliance for picking up and moving food items, in particular, in which two arms of the tong that can be locked together and subsequently unlocked with a single hand, without changing the grip on the tong, and without having to change the orientation of the tong, by utilizing a locking mechanism positioned on the arms of the tong in a manner that allows the lock to be actuated and released simply by the action of squeezing the arms together.

What is claimed is:

1. A tong, comprising:

- a) a first arm having a first proximal end, a first distal end opposite the first proximal end, a first inner surface in between the first proximal end and the first distal end, and a first outer surface opposite the first inner surface and in between the first proximal end and the first distal end, wherein the first distal end comprises a first flexible tip;
 - b) a second arm having a second proximal end, a second distal end opposite the second proximal end, a second inner surface in between the second proximal end and the second distal end, and a second outer surface opposite the second inner surface and in between the second proximal end and the second distal end, wherein the second distal end comprises a second flexible tip, wherein the first proximal end and the second proximal end are connected by a hinge;
 - c) a latch mounted on the first inner surface of the first arm; and
 - d) a strike mounted on the second inner surface of the second arm, wherein the latch and strike are configured to connect to each other in a locked configuration when the first flexible tip is compressed against the second flexible tip, and wherein the latch and strike are configured to disconnect from each other from the locked configuration when the first arm is squeezed towards the second arm, wherein in the locked configuration, a flanged head of the strike abuts against a post of the latch with a clasp of the latch surrounding the flanged head inside a housing of the latch, wherein in the locked configuration, the first distal end touches the second distal end, wherein in the locked configuration the latch and the strike are entirely interior to the first outer surface and the second outer surface, wherein the latch and the strike are fixed along the first arm and second arm, respectively, wherein the first inner surface defines a first channel, and the second inner surface defines a second channel, wherein the latch resides in the first channel, and the strike resides in the second channel, wherein the clasp of the latch protrudes out from the first channel, and the flanged head of the strike protrudes out from the second channel.
2. A tong, comprising:
- a) a first arm having a first proximal end, a first distal end opposite the first proximal end, a first inner surface in

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between the first proximal end and the first distal end, and a first outer surface opposite the first inner surface and in between the first proximal end and the first distal end;

- b) a second arm having a second proximal end, a second distal end opposite the second proximal end, a second inner surface in between the second proximal end and the second distal end, and a second outer surface opposite the second inner surface and in between the second proximal end and the second distal end, wherein the first proximal end and the second proximal end are connected by a hinge; and
- c) a lock configured to connect to the first arm and the second arm together when the first arm and the second arm are squeezed together, and wherein the lock is configured to disconnect the first arm and the second arm from each other when the first arm and the second arm are squeezed together, wherein the lock comprises a latch mounted on the first inner surface of the first arm and protruding away from the first arm towards the second arm; and a strike mounted on the second inner surface of the second arm and protruding away from the second arm towards the first arm, wherein the first distal end comprises a first flexible tip, and the second distal end comprises a second flexible tip, wherein the latch and the strike are configured to lock in a locked configuration when the first flexible tip is compressed against the second flexible tip, wherein in the locked configuration, the first distal end abuts against the second distal end, wherein the first inner surface defines a first channel, and the second inner surface defines a second channel, wherein the latch resides in the first channel, and the strike resides inside the second channel, wherein in the locked configuration the latch and the strike are entirely interior to the first outer surface and the second outer surface, wherein the latch and the strike are fixed along the first arm and second arm, respectively, wherein a portion of the latch protrudes out from the first channel, and a portion of the strike protrudes out from the second channel, and

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wherein the latch comprises a housing and a spring-loaded catch mounted inside the housing, wherein the spring-loaded catch comprises a post and a clasp connected to the post, wherein the strike comprises a base attached to the second inner surface and a flanged head opposite the base, wherein the latch and strike are configured to connect to each other in a locked configuration when the first arm and the second arm are substantially parallel to each other, wherein in the locked configuration, the flanged head abuts against the post with the clasp surrounding the flanged head inside the housing.

3. A method of using a tong, comprising:

- a) providing a tong as recited in claim 1; and
- b) using one hand, squeezing the first arm and the second arms toward each other until the lock locks the first arm to the second arm in the locked configuration.

4. The method of claim 3, wherein squeezing the first arm and the second arms toward each other until the lock locks the first arm to the second arm causes the first flexible tip to compress against the second flexible tip to lock the first arm with the second arm.

5. The method of claim 3, further comprising squeezing the first arm and the second arm towards each other to unlock the first arm from the second arm.

6. A method of using a tong, comprising:

- a) providing a tong as recited in claim 2; and
- b) using one hand, squeezing the first arm and the second arms toward each other until the lock locks the first arm to the second arm in the locked configuration.

7. The method of claim 6, further comprising squeezing the first arm and the second arm towards each other to unlock the first arm from the second arm.

8. The method of claim 6, wherein squeezing the first arm and the second arms toward each other until the lock locks the first arm to the second arm causes the first flexible tip to compress against the second flexible tip to lock the first arm with the second arm.

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